

## 136. OPTIMIZING HIT AEROBIC WORKOUTS: THE DECREASING LENGTH INTERVALS

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Knowledgeiswatt

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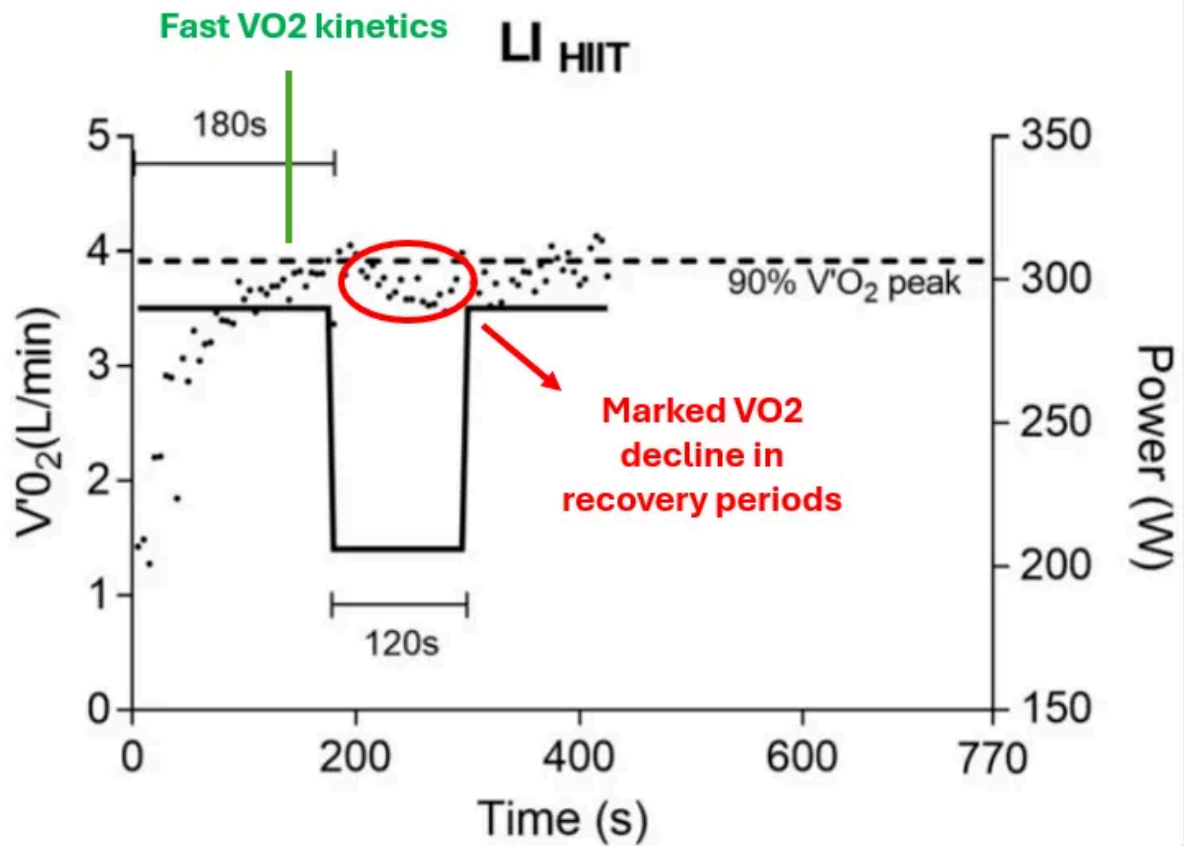


The search for optimized HIT sessions is based on understanding exercise physiology principles and how to exploit them by manipulating intensity, work, and recovery to maximize the training stimulus.

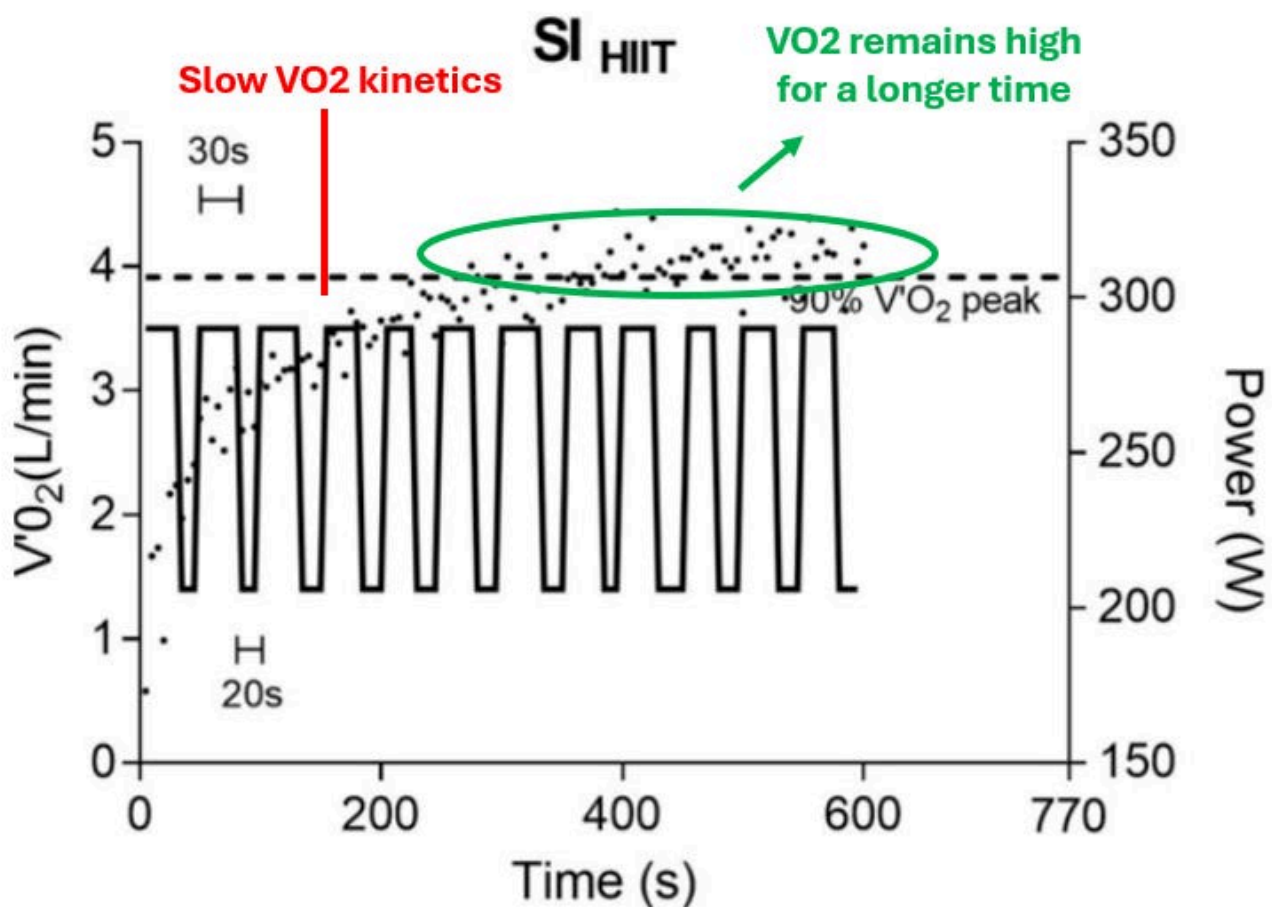
In this scenario, [in one of our most read KIW Research Note](#) we saw that **when the goal is to optimize the aerobic adaptations from high intensity (above the second threshold/Critical Power of FTP) training, the key is to accumulate as much time as possible at a high % of the maximum oxygen consumption (VO<sub>2</sub>max).**

Regarding this, we saw that **the short high intensity intermittent intervals** (series of 8-13 min with 3-5 min recovery between where you alternate 30-40s 'on' and 20-40s 'off' in a 1:1 or 2:1 work-recovery ratio) **are better than traditional long HIT intervals** (3-4 minutes work phases with 1:1 or 2:1 work-recovery ratio). It seems like the capacity to work at a higher power and fraction of VO<sub>2</sub>max despite a similar rate of perceived exertion is what make short intervals superior in promoting aerobic adaptations. One contributing factor is that during long intervals, the longer recovery periods—required to sustain the subsequent long work bouts—allow oxygen uptake (VO<sub>2</sub>) to drop substantially. As a consequence, the initial portion of each new long interval is spent rebuilding VO<sub>2</sub> rather than working at a high percentage of VO<sub>2</sub>max.

This results in a lower time spent at a high fraction of  $\text{VO}_2\text{max}$  across the entire session. In contrast, the short recovery periods typical of short intermittent intervals prevent  $\text{VO}_2$  from declining markedly. Oxygen uptake remains elevated throughout the session, leading to more time accumulated at a high percentage of  $\text{VO}_2\text{max}$ .



Oxygen consumption ( $\text{VO}_2$ ) during LONG HIT INTERVALS. Image modified from Vaccari et al. (1).

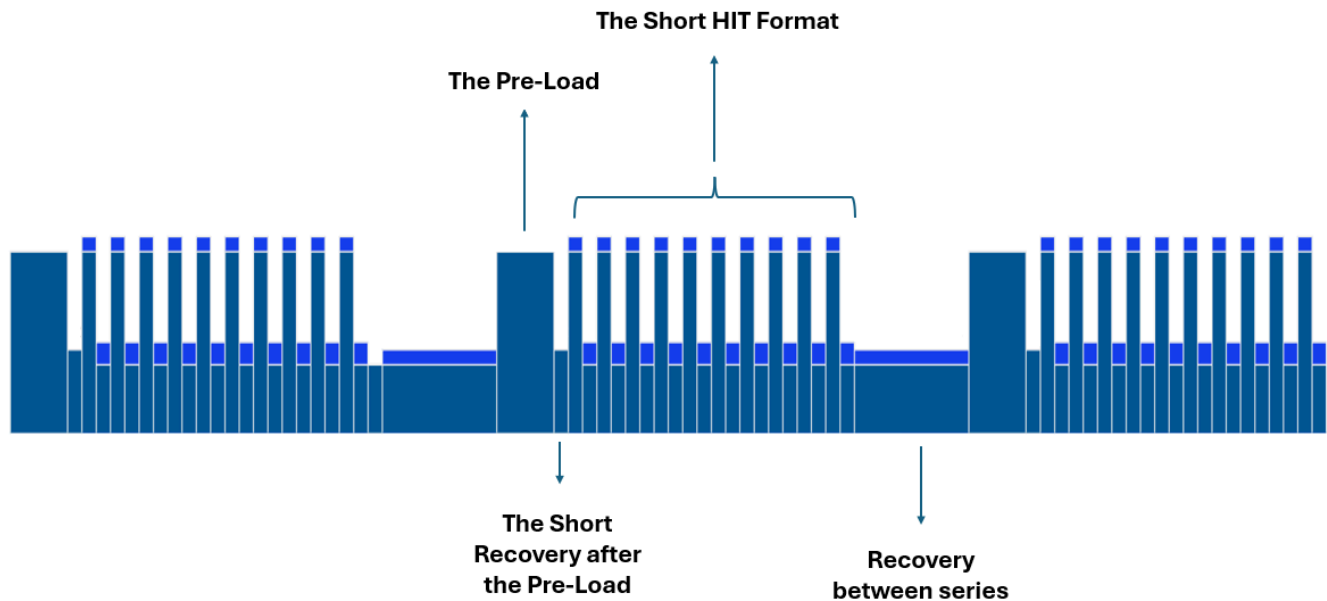


Oxygen consumption ( $\dot{V}O_2$ ) during SHORT INTERMITTENT INTERVALS. Image modified from Vaccari et al. (1).

Taking the concept one step further, in [the latest KIWAplied](#) we explored a creative combination of long and short intervals designed to further optimize the time spent at a high percentage of  $\dot{V}O_{2\max}$ , and therefore enhancing aerobic adaptations:

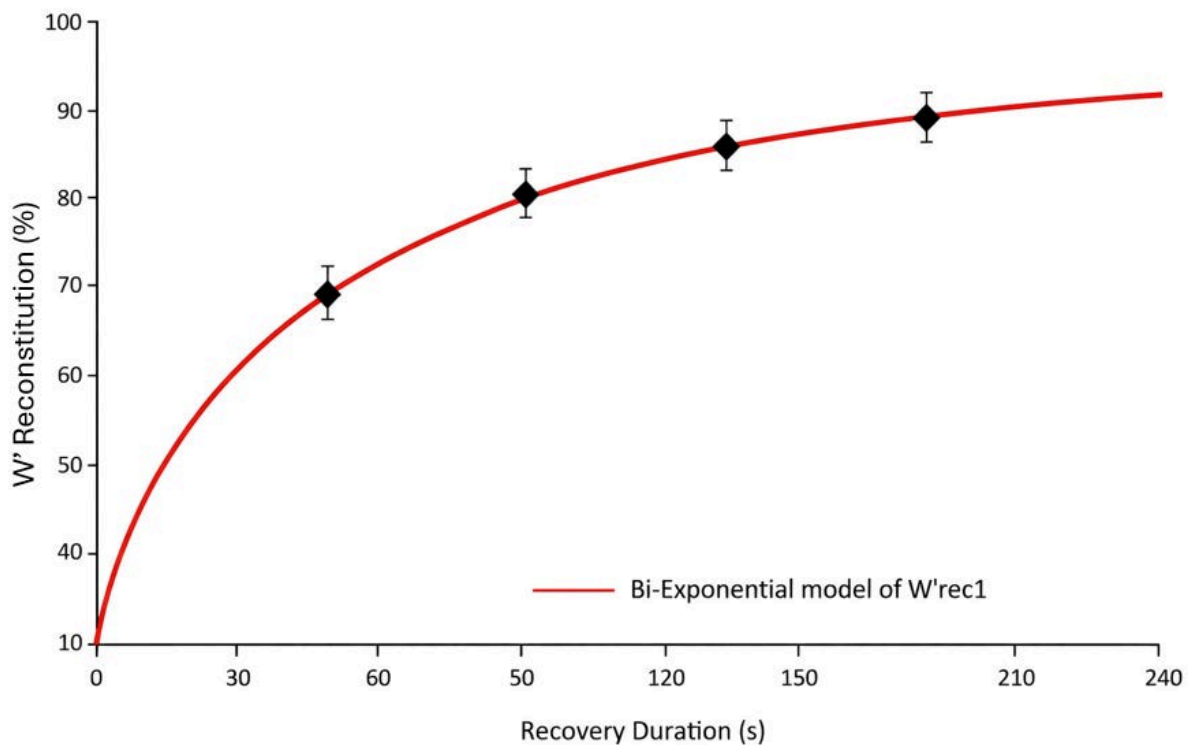
### The Pre-Loaded Short Intermittent Intervals.

This format combines long and short intermittent intervals within each series. How? Each series starts with a long interval (2–4 minutes, the '*pre-load*'), followed by a short recovery period (30s–1min), and then continues with 8–10 minutes of intermittent work. With this structure, athletes can exploit the faster  $\dot{V}O_2$  kinetics at exercise onset induced by the long interval, while simultaneously maximizing both the average time spent at a high percentage of  $\dot{V}O_{2\max}$  and the accumulated time above 90% of  $\dot{V}O_{2\max}$ , which are typically associated with short-intermittent intervals.



How is possible to go from a the pre-load long interval straight into the short intermittent intervals series with just a short (30s-1min) interval?

**Another key physiological principle which could be relevant trying to maximise the aerobic effectiveness of a HIT session relates to recovery kinetics. Following work performed above critical power (approximately FTP/second threshold), recovery processes—such as  $H^+$  removal, phosphocreatine (PCr) resynthesis, and so the reconstitution of the anaerobic reserve ( $W'$ )—follow an exponential time course. In practical terms, this means that the capacity to produce high-intensity work is restored very rapidly immediately after the end of an effort, while the rate of recovery progressively slows as recovery time increases.**



The exponential time course of W' reconstitution. Image modified from Cholely et al. 2021 (2).

How can this be applied in practice?

**Using a HIT format with *decreasing length intervals*. This approach could allow athletes to accumulate more time at a high % of  $\dot{V}O_{2\max}$ , thereby enhancing the aerobic stimulus of the session. How?**

- Starting with a **long first interval** (typically 2–4 minutes) which helps oxygen consumption rise rapidly toward near-maximal values.
- Once a high  $\dot{V}O_2$  is reached, decreasing length work intervals with decreasing length recovery period make it possible to have enough restoration to complete the next shorter effort while preventing  $\dot{V}O_2$  from dropping substantially.

***Does decreasing length HIT intervals optimize the time spent at high % of  $VO_{2\max}$  (and so likely aerobic adaptations) when compared with traditional short intermittent and long HIT intervals?***

A study published by Vaccari and Colleagues (Department of Medicine, University of Udine) on *European Journal of Applied Physiology* in 2020 tried to answer this question. (1)

## WHAT DID THEY DO?

- They recruited 12 amateur cyclists (average  $\dot{V}O_{2\max}$  57, Critical Power range 212–320 W).

- In the lab, participants completed three different HIT sessions on separate days.

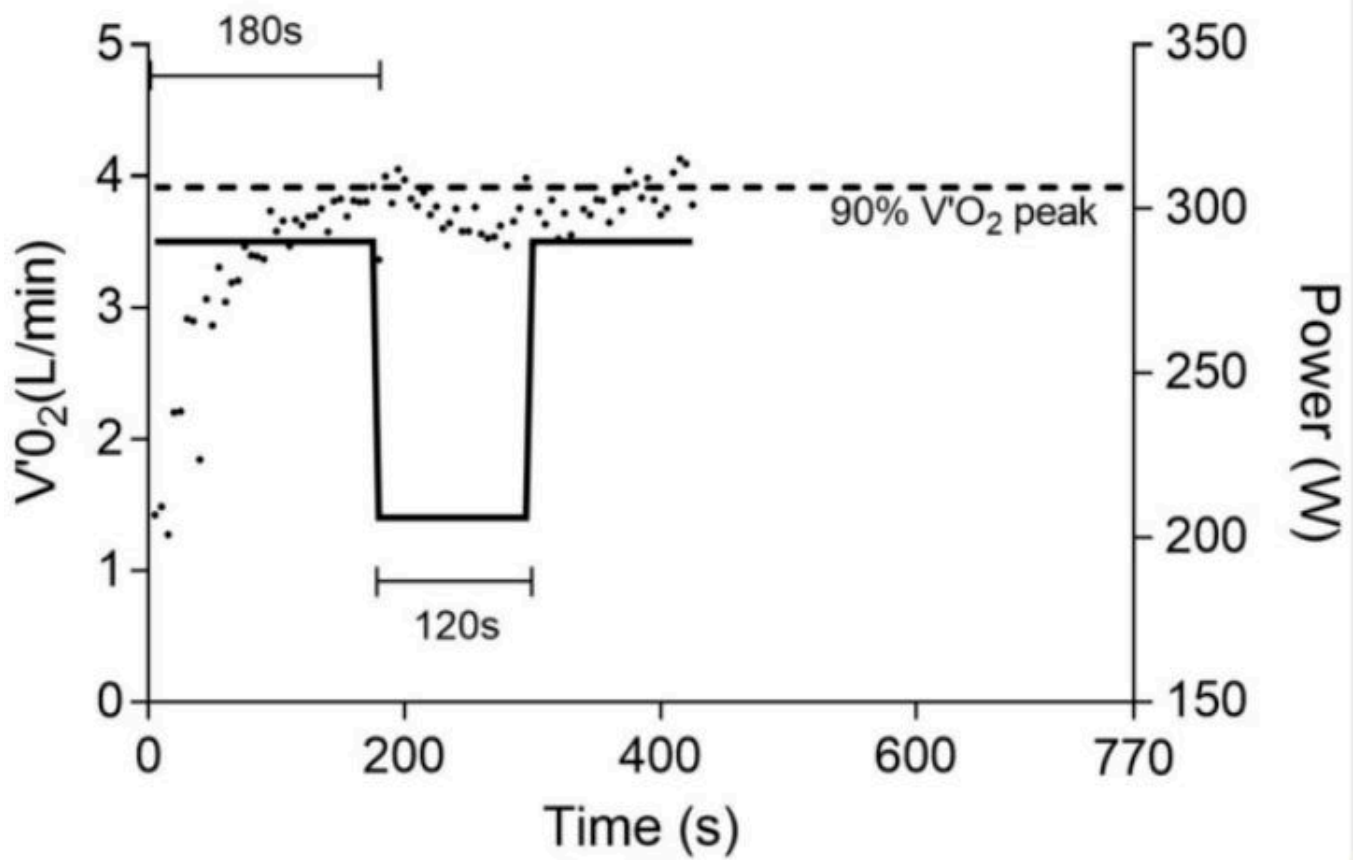
The three HIT workouts were structured as follows:

**Long Intervals ( $LI_{HIT}$ ):** 3 min at high intensity and 2 min recovery at low intensity repeated until exhaustion.

- **Short intervals ( $SI_{HIT}$ ):** 30 s at high intensity and 20 s recovery at low intensity repeated until exhaustion.
- **Decreasing Length Intervals ( $HIDIT$ ):**
  - 3 min at high intensity and 2 min recovery at low intensity;
  - 2 min at high intensity and 1min 20s recovery at low intensity;
  - 1 min at high intensity and 40 s recovery at low intensity;
  - 45 s at high intensity and 30 s recovery at low intensity;
  - finally 30 s at high intensity and 20 s recovery at low intensity, repeated until exhaustion.

You can see that in all sessions, the work-recovery ratio was always 3:2. In all the intervals of all the sessions the work phase was performed at around 117% of Critical Power (CP), while the recovery period at approximately 83% of CP.

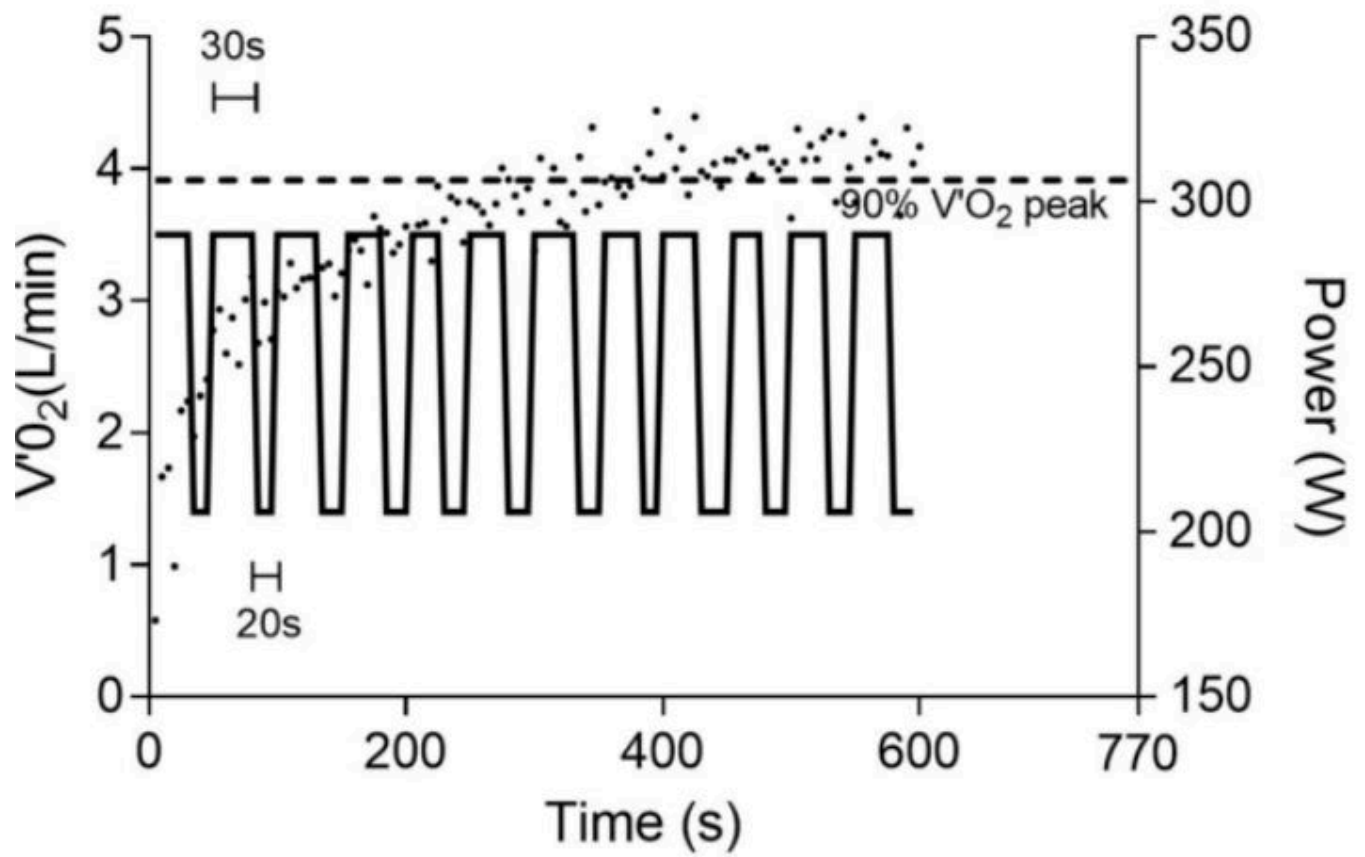
## LI HIIT



Long HIT Intervals. Modified from Vaccari et al. (1).

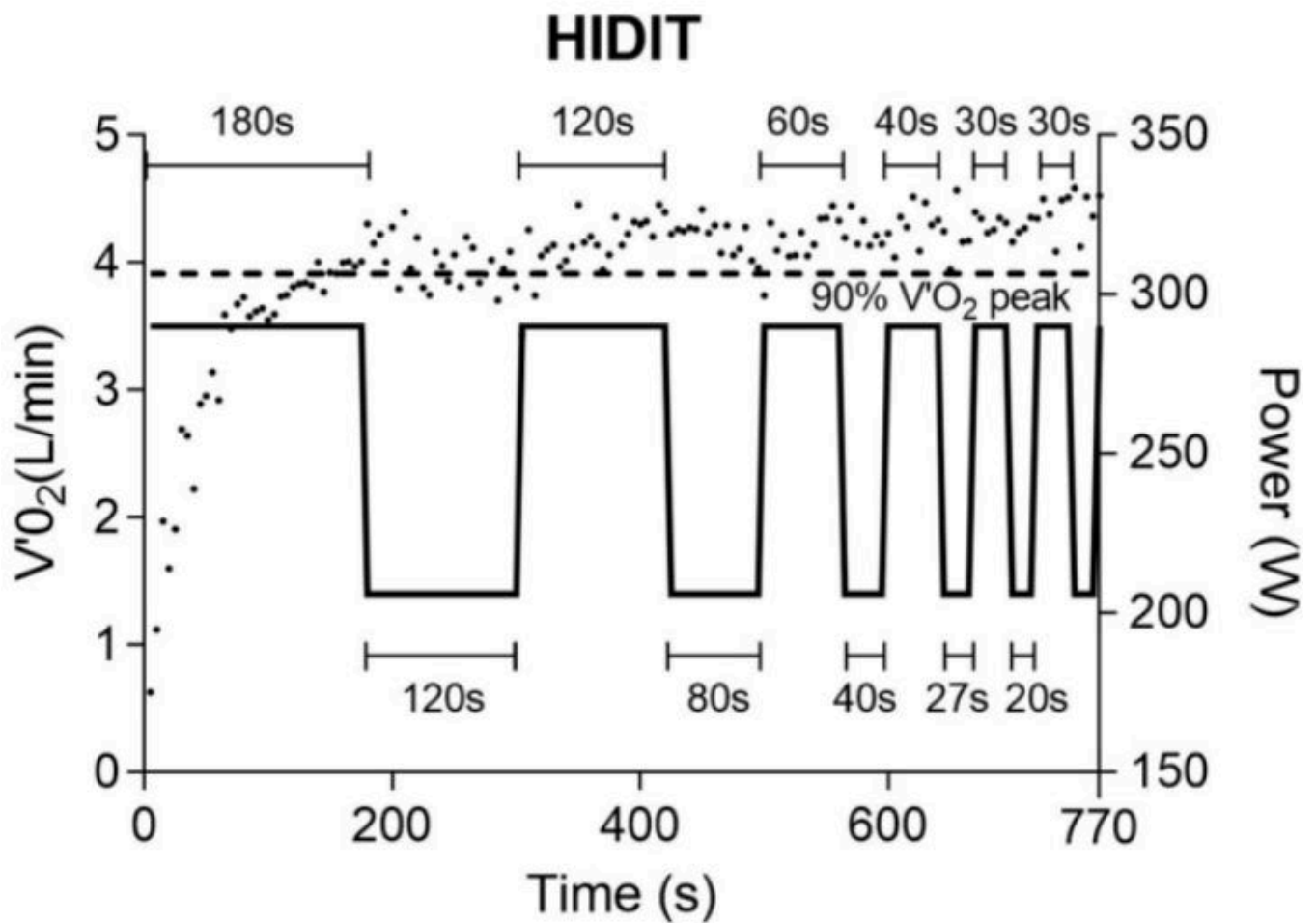


## SI HIIT



Short Intermittent Intervals. Modified from Vaccari et al. (1).





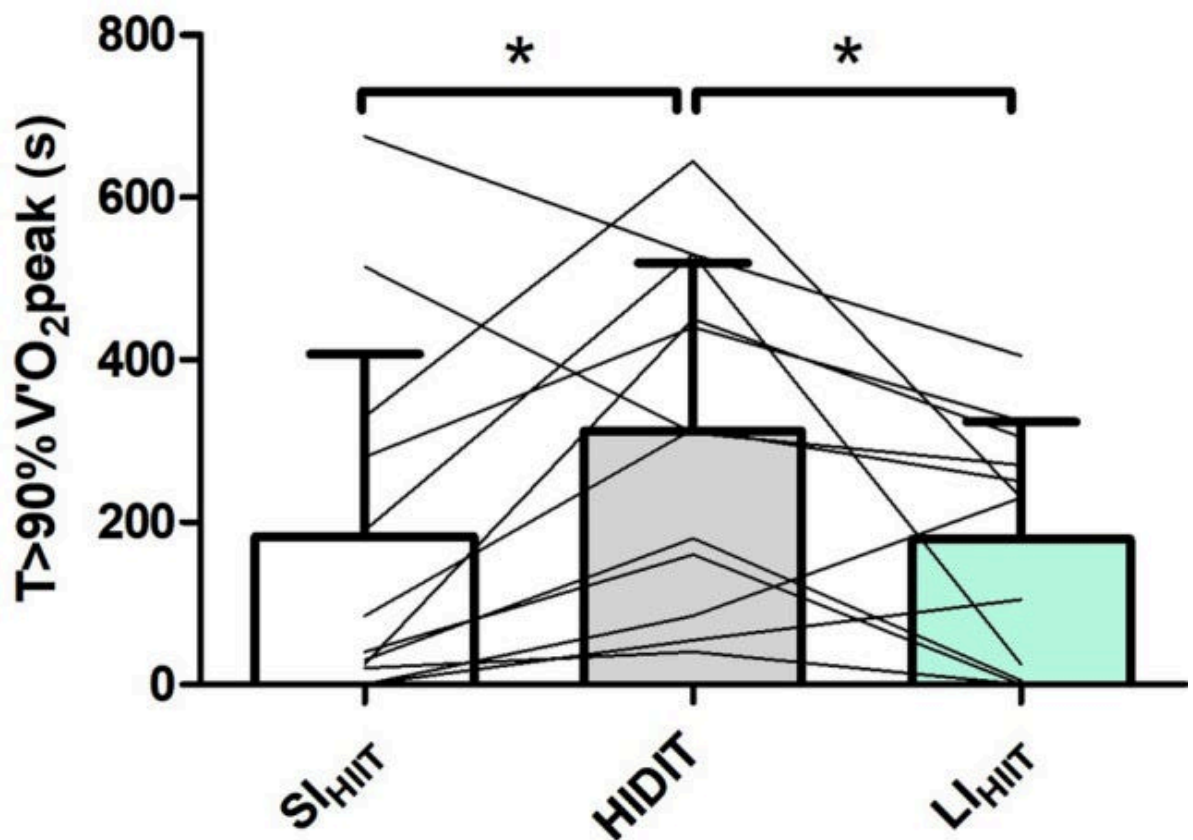
Decreasing Length Intervals. Modified from Vaccari et al. (1).

## WHAT DID THEY FIND?

The total time to exhaustion was higher in decreasing length compared to short intermittent and long HIT intervals:

- Decreasing Length:  $798 \pm 185$  s
- Short Intermittent:  $714 \pm 265$  s
- Long:  $664 \pm 282$  s

- The time spent above 90% of VO<sub>2</sub>max resulted higher in the decreasing length compared to the short intermittent and long HIT intervals:
  - Decreasing length: 312 ± 207 s
  - Short Intermittent: 183 ± 225 s
  - Long: 179 ± 145 s



Time spent above 90% of VO<sub>2</sub>max during the different HIT formats. Image from Vaccari et al. (1).

- The average fraction (%) of VO<sub>2</sub>max recorded during the decreasing length format resulted higher when compared with short intermittent and long HIT formats:
  - Decreasing length: 84. ± 4%
  - Short Intermittent: 82 ± 7%
  - Long: 80 ± 7%

## PRACTICAL TAKEAWAYS:

The Decreasing Length HIT format permits to maintain a high fraction of VO<sub>2</sub>max for a longer time when compared with the short intermittent and long intervals HIT formats. So, it could be an optimal solution when trying to optimize the aerobic adaptations deriving from HIT sessions.

## GABRIELE'S COMMENT

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- There is not much more to add. Taken together, these data suggest that **HIT formats with progressively shorter intervals** can be an effective strategy when the goal is to optimise aerobic adaptations. This is because **they exploit the exponential reconstitution of anaerobic energy reserves following high-intensity work, allowing athletes to sustain a higher fraction of VO<sub>2</sub>max for a longer time.**
- As always when reading a KIW Research Note, the suggestion is **not** to blindly copy and paste a protocol into your training plan. Instead, take home the *principle* and adapt it to your context using your experience and intuition. For example, regarding these data, I would **lower the recovery intensity** (zone 1 or 2 in a 5/7 zones model) and **avoid going to complete exhaustion within a single set**, and instead perform **multiple sets** (three) without reaching true failure (maybe only in the last one). I would also extend the sequence toward **even shorter intervals at the end**, potentially finishing with formats such as 12-8 sec.

**Take the principle, experiment, and build.**

I think decreasing-length intervals can be particularly effective during phases focused on physiological engine development, while may be less appropriate in the final build-up toward the goal competition, where I would put greater emphasis on race-specific high-intensity efforts.

Thanks for Reading! Share with your Team!

**Gabriele Gallo, PhD in Exercise and Sport Science**

## REFERENCE:

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1. Vaccari F, Giovanelli N, Lazzer S. High-intensity decreasing interval training (HIDIT) increases time above 90% VO<sub>2</sub>peak. Eur J Appl Physiol. 2020 Nov;120(11):2397-2405.
2. Chorley A, Bott RP, Marwood S, Lamb KL. Bi-exponential modelling of W' reconstitution kinetics in trained cyclists. Eur J Appl Physiol. 2022 Mar;122(3):677-689. doi: 10.1007/s00421-021-04874-3. Epub 2021 Dec 18. PMID: 34921345; PMCID: PMC8854279.





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